



Volume of cylinders

Task A: Calculating the volume of a cylinder

1) This is the formula for volume of a cylinder = $\pi r^2 h$, where r is the radius and h is the height. Which of the following are equivalent?

a $V = (\pi r)^2 h$

e $V = \pi \times r \times r \times h$ ✓

b $V = \pi h r^2$ ✓

f $V = \pi \times h \times r \times r$ ✓

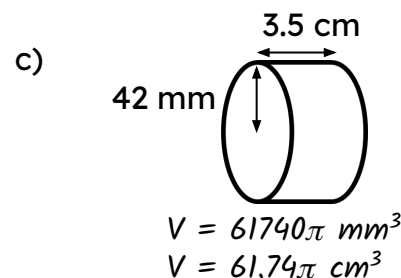
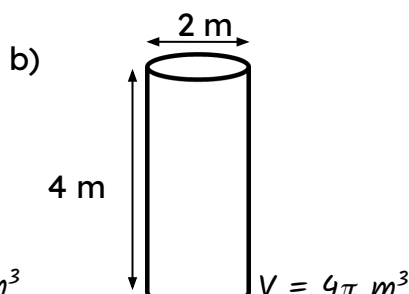
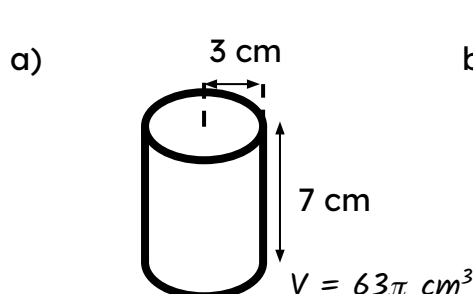
c $V = \pi r h^2$

g $V = \pi^2 r h$

d $V = \pi (r h)^2$

h $V = r^2 h \pi$ ✓

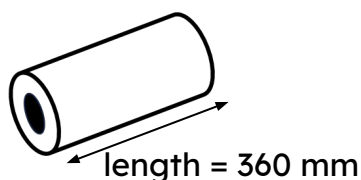
2) Calculate the volume of the following cylinders. Leave in terms of π .



d) A cylinder with a radius of 8 mm and a depth of 12 mm $V = 768\pi \text{ mm}^3$

e) A cylinder with a diameter of 4.8 cm and a height of 12 cm $V = 69.12\pi \text{ cm}^3$

3) A metal washer factory forms a long metal pipe before slicing them into individual washers. What volume of metal is used to create this section of pipe?



internal diameter = 12 mm
external diameter = 35 mm

Calculating in mm

$$V = (\pi \times 17.5^2 - \pi \times 6^2) \times 360$$

$$V = 97290\pi \text{ mm}^3 = 305645.5 \text{ mm}^3$$

Calculating in cm

$$V = (\pi \times 1.75^2 - \pi \times 0.6^2) \times 36$$

$$V = 97.29\pi \text{ cm}^3 = 305.6 \text{ cm}^3$$



Task B: Using the volume of a cylinder

1) All of these cylinders have the same volume. Calculate the missing lengths.

$$\text{Volume} = 360\pi \text{ cm}^3$$

